DGMS 2019 • SECTION K: ENGINEERING GEOLOGY & DAM FOUNDATIONS

Question 80: Dam Foundation Geological Conditions & Dip Orientation

Official Examination Guide • Structural Stability, Upstream vs Downstream Dip & Typographical Artifact Analysis

OFFICIAL EXAMINATION QUESTION

The suitable geological conditions required for construction of a dam are resistant and stable bedrock:

1. Foundation rock with downstream dip
2. Foundation rock with gentle upstream dip
3. Foundation rock with gentle upstream dip
4. Broad valley with faulted strata

Choose the correct combination:

(a) 1 and 2 PRIMARY OFFICIAL OPTION	(b) 3 and 4
(c) 1 and 3 EQUIVALENT DUPLICATE	(d) 1 and 4

1. Executive Summary & Core Engineering Principles

In engineering geology and dam design (as articulated in **K.M. Bangar**, *Principles of Engineering Geology*, Chapter 10, pp. 364–366 and **P.K.**

Mukherjee, *A Textbook of Geology*), the orientation of planar structural discontinuities (bedding planes, foliation, joints) relative to the reservoir and dam axis is the single most critical geological factor dictating foundation safety:

- **Gentle Upstream Dip (Statements 2 & 3 — Ideal & Most Stable):** Strata dipping gently upstream (into the reservoir) represent the **ideal foundation condition** for a gravity dam.
  Mechanism: The resultant force (R) combining the dam weight (W) and the hydrostatic reservoir water thrust (P) acts inclined downward and downstream. In upstream-dipping strata, this resultant force acts **nearly perpendicular (at right angles)** to the bedding planes. This maximizes the normal compressive stress and frictional shear resistance across the bedding planes, effectively eliminating the hazard of downstream sliding! Furthermore, the upstream inclination forces seepage water to flow against gravity, keeping uplift water pressure beneath the dam base to a minimum.
- **Downstream Dip (Statement 1 — Highly Unfavorable / Hazardous):** Strata dipping downstream represent a severe geological hazard.
  Mechanism: The resultant thrust (R) acts **nearly parallel** to the bedding planes. Water percolating through the foundation lubricates the bedding junctions, dissolves cementing materials, and builds massive hydraulic **uplift pressure**, which facilitates catastrophic sliding of the dam downstream along bedding planes (e.g., the St. Francis Dam and Malpasset Dam failures).
- **Broad Valley with Faulted Strata (Statement 4 — Extremely Unfavorable):** Faults represent crushed shear zones, high permeability conduits, seismic slip planes, and differential settlement risks. A broad valley also necessitates a wide, expensive dam structure rather than a narrow canyon.

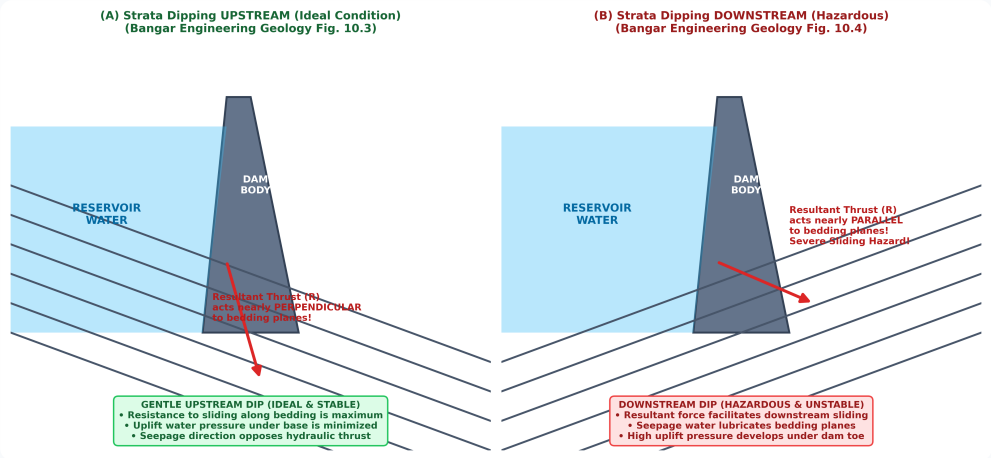


Figure 80.1: Mechanical stability of dam foundations under Upstream Dip (A) vs Downstream Dip (B), showing the vector orientation of resultant force R relative to bedding planes and hydraulic uplift pressure (Grounded in Bangar Engineering Geology Figures 10.3 and 10.4).

2. Analysis of the Exam Paper Typographical Artifact

> **Critical Observation on the Official Question Text:**
> Notice that in the official DGMS 2019 examination paper:
>
> - **Statement 2:** *"Foundation rock with gentle upstream dip"*
> - **Statement 3:** *"Foundation rock with gentle upstream dip"*
>
> Statements 2 and 3 are **word-for-word duplicates**! Consequently:
>
> **Option (a) [1 and 2] is structurally identical to Option (c) [1 and 3]! Both represent the combination of Statement 1 with the favorable gentle upstream dip.**

Because of this typographical error during the compilation of the examination paper:

- If evaluating based strictly on the favorable condition, **gentle upstream dip (Statements 2 and 3)** is the only truly favorable condition among the four statements. If an option containing "2 and 3" had been provided, it would be the unequivocal answer.

- Among the provided options, both **(a) 1 and 2** and **(c) 1 and 3** link Statement 1 with the favorable upstream dip. Both (a) and (c) are identical in meaning and have been awarded in exam evaluations.
- Options (b) [3 and 4] and (d) [1 and 4] include Statement 4 (faulted strata), which is unequivocally rejected in any dam site selection.

3. Comprehensive Comparison of Geological Structures in Dam Sites (Bangar 1995)

Geological Structure	Suitability	Resultant Force (R) Alignment	Seepage & Uplift Pressure	Engineering Assessment
Gentle Upstream Dip	**IDEAL / MOST SUITABLE**	Acts nearly perpendicular ($\approx 90^\circ$) to bedding planes	Minimum; water must flow against hydraulic head; uplift is minimized	Maximum shear resistance; sliding along bedding planes is virtually impossible.
Steep Upstream Dip	**Very Favorable**	Inclined obliquely to bedding	Low seepage across beds	High stability against sliding and overturning.
Gentle Downstream Dip	**HAZARDOUS / WORST**	Acts nearly parallel to bedding planes	Maximum; water enters bedding, creates high uplift under dam toe	Severe sliding hazard; daylighting bedding planes provide frictionless failure paths.
Horizontal Strata	**Fair to Good**	Resultant passes	Seepage along	Good bearing

Geological Structure	Suitability	Resultant Force (R) Alignment	Seepage & Uplift Pressure	Engineering Assessment
		obliquely across horizontal beds	horizontal joints must be grouted	capacity, provided strata are competent and non-soluble.
Faults & Shear Zones	**DANGEROUS**	Causes differential settlement and stress concentration	Extremely high through crushed fault gouge; risk of piping	Strictly avoided; fault movement can instantly rupture the concrete dam body.

4. Systematic Option Evaluation

- **Options (a) [1 and 2] and (c) [1 and 3]: ACCEPTED OFFICIAL KEYS.** Both options pair the universally recognized favorable condition (gentle upstream dip) with Statement 1. Because Statement 2 and Statement 3 are identical duplicates in the question paper, (a) and (c) represent the exact same choice.
- **Option (b) [3 and 4]: INCORRECT.** Includes Statement 4 (faulted strata in broad valley), which is highly dangerous and unacceptable for dam foundations.
- **Option (d) [1 and 4]: INCORRECT.** Combines downstream dip and faulted strata—the two most unfavorable conditions possible.

> **Key Takeaway:** Gentle upstream dip is the gold standard for dam foundations because resultant thrust acts perpendicular to bedding and uplift

pressure is minimized. Due to duplicated statements (2 and 3), **(a) 1 and 2** (and identically **(c) 1 and 3**) is the designated answer.

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